Lecture 25

3/18/26

Housekeeping

- Take-home midterm is available
  - Pick up hard copy at end of lecture
  - Download copy from Moodle
- MT has three quantitative problems that are similar to those on practice MT
- Key to practice MT is on Moodle
- MT deadline = Friday, April 3rd, at 5pm

ACTIVITY 8

Did Problem 6 - 11 (the remaining problems)

Ultrasound (continued)

Echo-based imaging

US imaging uses a device called a transducer that both creates & detects US

- Transducer sends an US pulse into patient
- Pulse is partially reflected, partially transmitted at each interface
- Measure go-return times
- Calculate depths from kinematic equation

$$2D = V_{us} \cdot t_{echo}$$

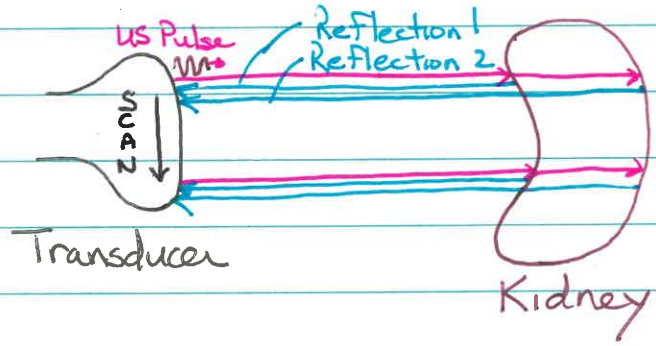
$$\boxed{D = \frac{V_{us}}{2} \cdot t_{echo}}$$

$V_{us} = 1540\ m/s$

Measured

Depth relative to transducer

SIMULATION



- Send series of pulses
- Detect series of sets of reflections (from each interface)
- Calculate series of Ds

Build up contours in image by scanning through sample

Assign locations based on echo time
Assign brightness based on echo strength

Strong echo → Bright/White
Intermediate echo → Variable shades of gray
No echo → Dark/Black

US imaging has a variety of strengths & weaknesses:

Strengths	Weaknesses
Very fast - real time	
Relatively inexpensive	
Portable	
Safe	

Ultrasound Frequencies

Diagnostic US has frequencies, ν, in a range from ~ 1 MHz to ~ 20 MHz

This is about 1000 times the frequency range for hearing

Why does US imaging employ a range of frequencies?

Huge Punchline for US frequencies

- Use HIGH frequencies to achieve the best resolution

  High frequency ⇒ Short wavelength ⇒ BEST resolution

- Use LOW frequencies to get deeper penetration because high frequency US is dissipated more quickly & so doesn't go as deep

Bottom Line
- Surface structures/thin samples (e.g., carotid, mammogram) ⇒ use high ν
- Deep structures (e.g., organs like liver) ⇒ use low ν

Why use frequencies in the MHz regime?

The resolution goal in medical imaging is ~1mm

0.1 mm to 20 mm (2 cm)
- 0.1 mm: x-rays
- 20 mm: radionuclide

Mammography is the highest resolution medical imaging technique

$\boxed{\text{Resolution} \approx \lambda/2}$ Rule of Thumb

Need $\lambda \approx 1\,mm$ for Resolution $= 0.5\,mm$

For waves, v, λ, ν are related by the equation

$$\boxed{v = \lambda \nu}$$

v = wave speed = 1540 m/s
λ = wavelength
ν = frequency

The frequency required for 0.5 mm resolution is 1.5 MHz

$$\nu = \frac{v}{\lambda} = \frac{1540\,m/s}{1 \cdot 10^{-3}\,m} = 1.54\,MHz \approx 1.5\,MHz$$

Thus, need MHz US to get mm resolution

Quantifying Contrast

We know, from above, how to use echo ranging to map feature depths & learn shapes

How do you get contrast in US images?

Get contrast from strength & nature of reflections

Often get impression that US bounces off a smooth structure, but this is not the case

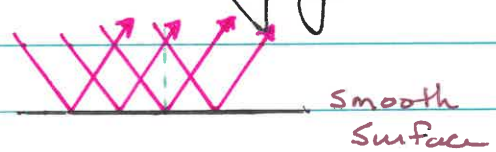
Most body surfaces are rough & create diffuse reflection

Medical imaging is all about interactions of waves w/ matter

Ultrasound Interactions w/ Body

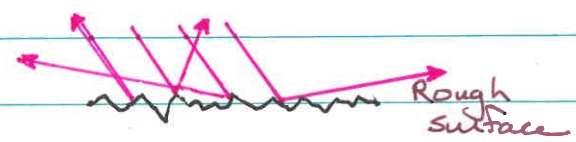
There are three ways US interacts with the body that are of interest in US imaging

(1) Specular Reflection

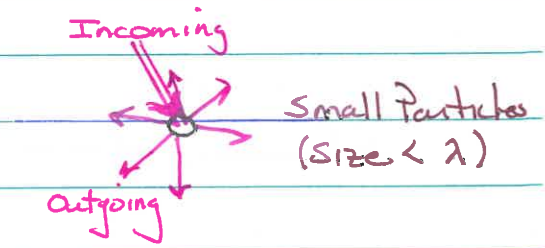


Reflection off smooth surface
Easiest to quantify
Example: the diaphragm

(2) Diffuse Reflection



Reflection off rough surface
Multi-directional → hard to quantify
More common than specular reflection



(3) Scattering

Reflection off small features
(e.g., red blood cells), smaller than wavelength of US
Multi-directional

Doctor is NOT always interested in quantifying interactions, but can instead interpret image based on appearance
Doctor know what's normal & what's not (& why)

Quantifying Specular Reflection

Specular reflection can be quantified using a "reflection coefficient" defined as the ratio of the reflected intensity to the incident intensity:

$$R = \frac{I_r}{I_i} = \frac{\text{Reflected Intensity}}{\text{Incident Intensity}}$$

This expression can be rewritten in terms of something called the "acoustic impedance", z

Acoustic impedance is the sound analog of the refractive index, n, used w/ light

z measures resistance to motion, like n

$$R = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

Interface: Med 1 Z_1 / Med 2 Z_2

This equation is an exact analog of the equation that describes reflection at an optical interface, w/ "Z" replacing "n"

Look at three special cases

(1) Matched Interfaces $\quad Z_1 = Z_2 \Rightarrow R = 0$

There is no reflection at matched interfaces

This situation is rare

(2) Minimally Mismatched Interfaces $\begin{cases} Z_1 \approx Z_2 & \text{close} \\ R = 0.1 - 1\% \end{cases}$

There is minimal but still nonzero reflection at minimally mismatched interfaces

This situation is common w/ soft tissues
Looked at Z values for a variety of tissues
They were very close

(3) <u>VERY Mismatched Interfaces</u> $\left\{ \begin{array}{l} Z_1 \text{ \& } Z_2 \text{ VERY different} \\ R \text{ is BIG} \end{array} \right.$

There is significant (up to almost total) reflection at very mismatched interfaces

This situation applies to "air to skin" & to a slightly lesser extent to "soft tissue to bone"

Use a "coupling gel" positioned between the transducer & the skin to reduce mismatch and allow US to enter body

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Handed out take-home midterm, covering microscopy & w/ 4 simulations on US